

Agenda:

1. Truth Table for Bitwise Operators
 2. Basic AND, OR & XOR Properties
 3. Question: Single Number
 4. Left Shift & Right Shift Operators
 5. Set, Flip, Unset & Check Bit
 6. Question: Single Number 3 (2 unique elements)
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Bit-wise Operators : $\&$, $|$, $\wedge$, $\sim$, $\ll$, $\gg$

1 $\rightarrow$ set bit / on / true / high voltage

0 $\rightarrow$ unset bit / off / false / low voltage

		AND	OR	XOR	NOT
a	b	$a \& b$	$a b$	$a \wedge b$	$\sim a$
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

Some some puppy shame (pointing to XOR)

NOT !a (pointing to NOT)

$A = 45$ $45 \rightarrow$ 101101

$B = 20$ $20 \rightarrow$ 010100

$A \wedge B = ?$

$111001 \rightarrow 57$



Basic Properties

 $5 \rightarrow 101$
 $6 \rightarrow 110$
 $11 \rightarrow 1011$
 $12 \rightarrow 1100$

1. Even / Odd Number $\rightarrow$

 $11 \rightarrow 1 \ 0 \ 1 \ 1$
 $\& \quad 0 \ 0 \ 0 \ 1$
 $0 \ 0 \ 0 \ 1$
 $10 \rightarrow 1 \ 0 \ 1 \ 0$
 $\& \quad 0 \ 0 \ 0 \ 1$
 $0 \ 0 \ 0 \ 0$

$N \& 1 \rightarrow 0 \Rightarrow N \text{ is even}$
 $\quad \quad \quad \rightarrow 1 \Rightarrow N \text{ is odd}$

 $A = 5 \rightarrow 101$

2. $A \& 0 \rightarrow \underline{0}$

 101
 $\& \underline{000}$
 $\underline{000}$

3. $A \& A \rightarrow \underline{A}$

 101
 $\& \underline{101}$
 $\underline{101}$



4. $A | 0 \rightarrow \underline{A}$

$$\begin{array}{r} 101 \\ | 000 \\ \hline 101 \end{array}$$

5. $A | A \rightarrow \underline{A}$

$$\begin{array}{r} 101 \\ | 101 \\ \hline 101 \end{array}$$

6. $A \wedge 0 \rightarrow \underline{A}$

$$\begin{array}{r} 101 \\ ^\wedge 000 \\ \hline 101 \end{array}$$

7. $A \wedge A \rightarrow \underline{0}$

$$\begin{array}{r} 101 \\ ^\wedge 101 \\ \hline 000 \end{array}$$

**Commutative Property** →

$$a + b = b + a$$

$$a - b \neq b - a$$

$$A \& B = B \& A$$

$$A | B = B | A$$

$$A \wedge B = B \wedge A$$

Associative Property →

$$(a + b) + c = a + (b + c)$$

$$(A \& B) \& C = A \& (B \& C)$$

$$(A | B) | C = A | (B | C)$$

$$(A \wedge B) \wedge C = A \wedge (B \wedge C)$$

$$5 \rightarrow 101$$

$$6 \rightarrow 110$$

$$4 \rightarrow 100$$



< Question- 1 > : Evaluate the expression: $a \wedge b \wedge a \wedge d \wedge b$

$$= (a \wedge a) \wedge (b \wedge b) \wedge d$$

$$= (0 \wedge 0) \wedge d = 0 \wedge d = \underline{d}$$

< Question- 2 > : Evaluate the expression: $1 \wedge 3 \wedge 5 \wedge 3 \wedge 2 \wedge 1 \wedge 5$

$$= (1 \wedge 1) \wedge (3 \wedge 3) \wedge (5 \wedge 5) \wedge 2$$

$$= (0 \wedge 0) \wedge (0 \wedge 2)$$

$$= 0 \wedge 2 = \underline{2}$$

< Question > : Given arr[N] where every element is present twice except one unique element.
Find that unique element.

$$A = [5 \quad 6 \quad 8 \quad 5 \quad 6 \quad 7 \quad 7]$$

Ans = $\forall i \wedge A[i]$, XOR of A[i]

$$5 \wedge 6 \wedge 8 \wedge 5 \wedge 6 \wedge 7 \wedge 7$$

$$= (5 \wedge 5) \wedge (6 \wedge 6) \wedge (7 \wedge 7) \wedge 8$$

$$= 0 \wedge 0 \wedge 0 \wedge 8 = 0 \wedge 8 = \underline{8}$$

$$ans = A[0]$$

for $i \rightarrow 1$ to $(N-1)$ {

$$ans \wedge = A[i]$$

$$TC = \underline{O(N)} \quad SC = \underline{O(1)}$$

} return ans

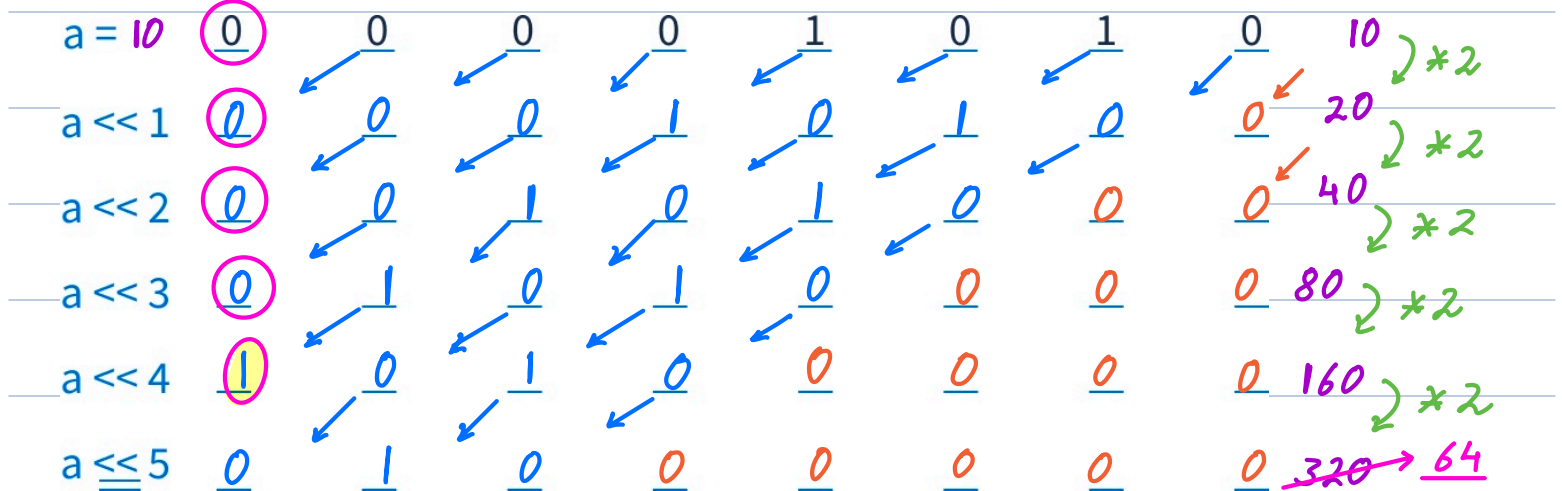


Left Shift Operator (<<)

int = 32 bits

lets use 8 bit system for our understanding.

discarded



(overflow)

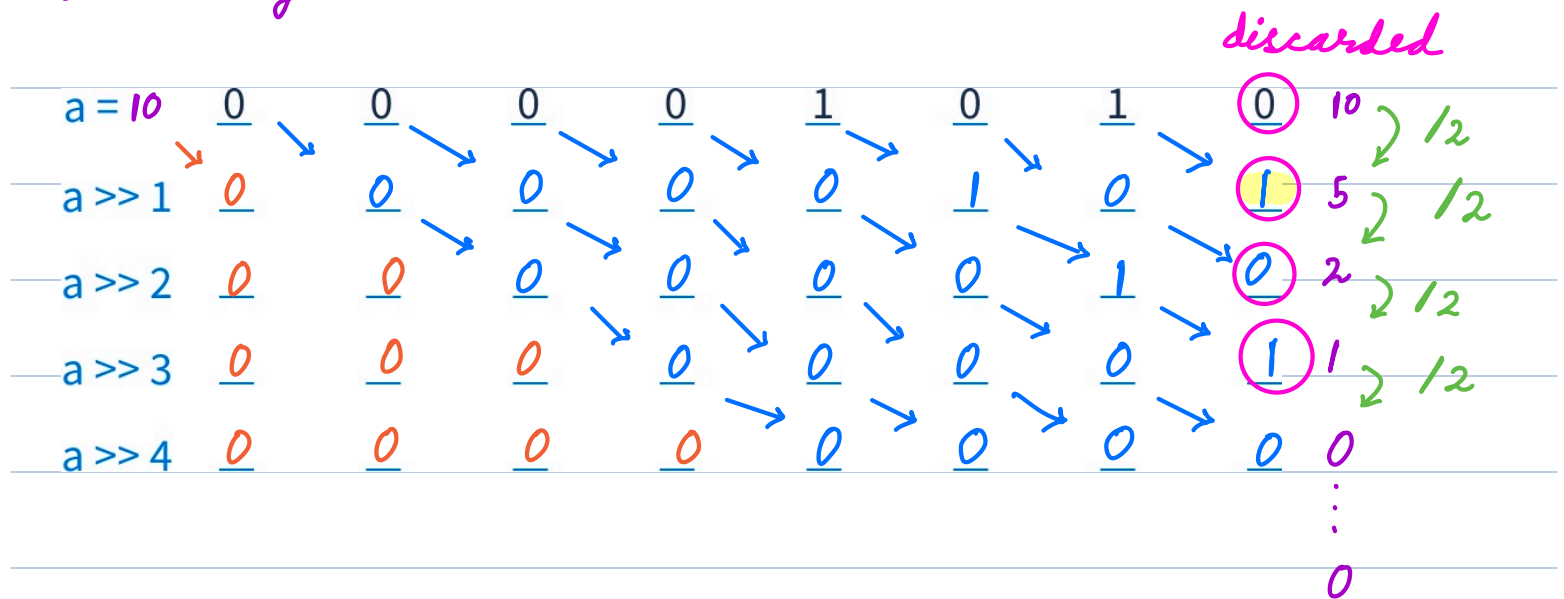
$$A \ll n = A * 2^n$$

$$1 \ll n = 2^n$$



Right Shift Operator (>>)

[8 bit system]



$$A \gg n = A / 2^n$$



Power of Left Shift Operator

$1 \ll i \rightarrow$ only i^{th} bit is set

1. OR Operator $\rightarrow$

$N = 45$

5	4	3	2	1	0
1	0	1	1	0	1

$1 \ll 2$

0	0	0	1	0	0
---	---	---	---	---	---

1 0 1 1 0 1 $\rightarrow$ 45

5	4	3	2	1	0
1	0	1	1	0	1

$1 \ll 4$

0	1	0	0	0	0
---	---	---	---	---	---

1 1 1 1 0 1 $\rightarrow 45 + 2^4$
 $= 45 + 16 =$ 61

$N | (1 \ll i) \rightarrow$ set i^{th} bit of N

3. XOR Operator $\rightarrow$

$N = 45$

5	4	3	2	1	0
1	0	1	1	0	1

$1 \ll 2$

0	0	0	1	0	0
---	---	---	---	---	---

1 0 1 0 0 1 $\rightarrow 45 - 2^2$

= 41

5	4	3	2	1	0
1	0	1	1	0	1

$1 \ll 4$

0	1	0	0	0	0
---	---	---	---	---	---

1 1 1 1 0 1 $\rightarrow$ 61

$N \wedge (1 \ll i) \rightarrow$ flip/toggle i^{th} bit



2. AND Operator →

$N = 45$ 5 4 3 2 1 0
 1 0 1 1 0 1

$1 \ll 2$ & 0 0 0 1 0 0

0 0 0 1 0 0 → $2^2 = 4$

5 4 3 2 1 0
 1 0 1 1 0 1

$1 \ll 4$ & 0 1 0 0 0 0

0 0 0 0 0 0 → 0

check
 i^{th} bit.

$N \& (1 \ll i) \rightarrow 0 \Rightarrow i^{\text{th}}$ bit is unset in N
 $(1 \ll i) \Rightarrow i^{\text{th}}$ bit is set in N

set

OR

Flip

XOR

check

&

unset



$\text{if } (A \& (1 \ll i) > 0)$

$A = A \wedge (1 \ll i)$



< Question > : Given an integer array of size N, where all elements repeat twice except two. Find those two elements.

arr[] $\rightarrow$ [4 , 5 , 4 , 1 , 6 , 6 , 5 , 2]

{ T.C - $O(N)$
S.C - $O(1)$ }

ans = A[0]

for i $\rightarrow$ 1 to (N-1) {

TC = $O(N)$ SC = $O(1)$

ans $\wedge$ = A[i]

} return ans

A = [5 4 5 3 4]

	2	1	0
5 $\rightarrow$	1	0	1
4 $\rightarrow$	1	0	0
5 $\rightarrow$	1	0	1
3 $\rightarrow$	0	1	1
4 $\rightarrow$	1	0	0

of set bit at i^{th} position

$\rightarrow$ even (divisible by 2)

$\rightarrow$ odd (not divisible by 2)

1's $\rightarrow$ 4 / 3

0 1 1 $\rightarrow$ 3

Problem Statement

Q →

Given an integer array, all the elements will occur **twice except two**. Find **those two elements**.

Input: [4, 5, 4, 1, 6, 6, 5, 2]

Output: 1, 2

4 → 100

5 → 101

6 → 110

$$4 \wedge 5 \wedge 4 \wedge 1 \wedge 6 \wedge 6 \wedge 5 \wedge 2 = 1 \wedge 2 = 3$$

$$x \wedge y > 0 \quad \text{if } x \neq y$$

3 2 1 0
→ 0 0 1 1

either x or y

(but not both) will have

1 → { 5 1 5 } → XOR of all → 1 this bit = 1.

0 → { 4 4 6 6 2 } → XOR of all = 2

xor = 0

for $i \rightarrow 0$ to $(N-1)$ {

| xor ^= A[i]

}

b = -1

for $i \rightarrow 0$ to 31 { // integer

| if ((xor & (1 << i)) > 0) {

```
        b = i
        break
    }
}
```

```
x = 0      y = 0
for i → 0 to (N-1) {
    if ( A[i] & (1 << b) == 0 ) x ^= A[i]
    else y ^= A[i]
}
```

```
return {x, y}
```

TC = $O(N)$ SC = $O(1)$
